Quantification: 682_z29
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	15
interior lacuna count	13
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 13)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1650.75	1511.5	948.377	px^2
major_axis_length_px	lacuna shape	74.0009	74.236	22.3194	px
minor_axis_length_px	lacuna shape	27.9394	29.9101	9.5919	px
aspect_ratio	lacuna shape	2.805	2.6218	0.775	unitless
eccentricity	lacuna shape	0.9144	0.9241	0.0615	unitless
solidity	lacuna shape	0.8593	0.8693	0.0551	unitless
roots_count	headline	8.2308	9	3.4194	count
ring_length_r30_px	headline	343.692	363	128.306	px
ring_length_r60_px	reported	824.077	938	287.943	px
owned_length_px	ownership-dependent	2440.89	1579.88	1838.77	px
edge_count	ownership-dependent	106.846	62	83.2715	count
mean_edge_length_px	ownership-dependent	23.4937	23.5198	2.0628	px
perimeter_px	normalised	186.054	202.693	70.3766	px
ring_area_r30_px2	normalised	7571.77	7711	2170.77	px^2
ring_area_r60_px2	normalised	19073	19480	4883.71	px^2
in_frame_fraction_r30	normalised	0.9952	1	0.0173	unitless
in_frame_fraction_r60	normalised	0.9839	1	0.0537	unitless
ring_density_r30	normalised	0.0453797	0.0442874	0.0106587	px^-1
ring_density_r60	normalised	0.0429315	0.0434665	0.00848081	px^-1
roots_per_100px_perimeter	normalised	4.4574	4.6693	1.0284	per 100 px
ring_attached_length_r30_px	network v2	311.923	319	128.069	px
ring_attached_length_r60_px	network v2	709.769	739	295.661	px
ring_length_w_r30_px	network v2	386.788	410.635	141	px
ring_length_w_r60_px	network v2	929.157	1048.9	323.319	px
sholl_crossings_r10	network v2	8.2308	8	3.0319	count
sholl_crossings_r20	network v2	10.5385	11	4.0541	count
sholl_crossings_r30	network v2	11.6923	11	3.9872	count
loop_count	loops, report only	10.3846	3	13.2636	count
loop_edge_share	loops, report only	0.2305	0.2256	0.2031	share
ring_width_mean_r30_px	width	6.9699	6.9555	0.6104	px

Per field
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measure	value	units
analysed_area_px2	1.02273e+06	px^2
total_skeleton_length_px	38267	px
canalicular_length_density_per_px	0.0374166	px^-1
junction_count	855	count
junction_density_per_px2	0.000836	px^-2
lacuna_count	15	count

lacunae_per_px2	1.467e-05	px^-2
skeleton_component_count	230	count
mean_component_length_px	166.378	px
median_component_length_px	21	px
field_length_density_w_per_px	0.0422627	px^-1
field_density_without_flagged_per_px	0.0377562	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	419	px
unattached_component_count	27	count
canalicular_length_density_attached_per_px	0.0374166	px^-1
field_length_density_with_unattached_per_px	0.0378263	px^-1
loop_count	172	count
loop_edge_share	0.3531	share
edge_count	1869	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.7082	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10.198	px
width_pixels_used	35757	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2083	86.508	34.896	2.479	0.915	0.8266	1.5435	13.4983	99.4023	True
2	2213	89.1614	32.1486	2.7734	0.9327	0.8778	-1.1291	134.914	921.292	False
3	446	49.6136	12.2658	4.0449	0.969	0.7866	-1.4445	249.987	449.359	False
4	3140	120.617	34.4194	3.5043	0.9584	0.8885	1.5609	431.048	220.093	False
5	2837	88.9183	41.496	2.1428	0.8844	0.9229	-1.5453	532.043	493.055	False
6	1634	75.9278	30.0551	2.5263	0.9183	0.8211	-1.4797	555.483	357.187	False
7	3781	121.501	43.6088	2.7862	0.9334	0.8771	1.4272	590.482	55.0987	True
8	1399	81.52	23.1918	3.515	0.9587	0.8541	1.5542	716.135	181.57	False
9	663	61.7115	15.8655	3.8897	0.9664	0.7408	-1.1603	728.596	861.06	False
10	513	41.8409	17.3844	2.4068	0.9096	0.8607	1.3405	830.565	75.5867	False
11	1130	64.3164	23.6688	2.7173	0.9298	0.8835	1.4062	842.292	125.237	False
12	1244	49.7173	33.3792	1.4895	0.7411	0.9325	-1.542	904.6	760.807	False
13	2966	92.1224	41.6334	2.2127	0.8921	0.895	-1.4899	920.401	307.385	False
14	1624	72.5441	29.7651	2.4372	0.9119	0.8476	-0.945	978.497	437.615	False
15								548.08	212.663	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	4	192	359	1753.54	96	18.2661	
2	14	556	1343	6503.61	282	23.0624	6.9555
3	6	363	942	4290.4	197	21.7787	7.2279
4	11	459	1235	3695.9	192	19.2495	7.4568

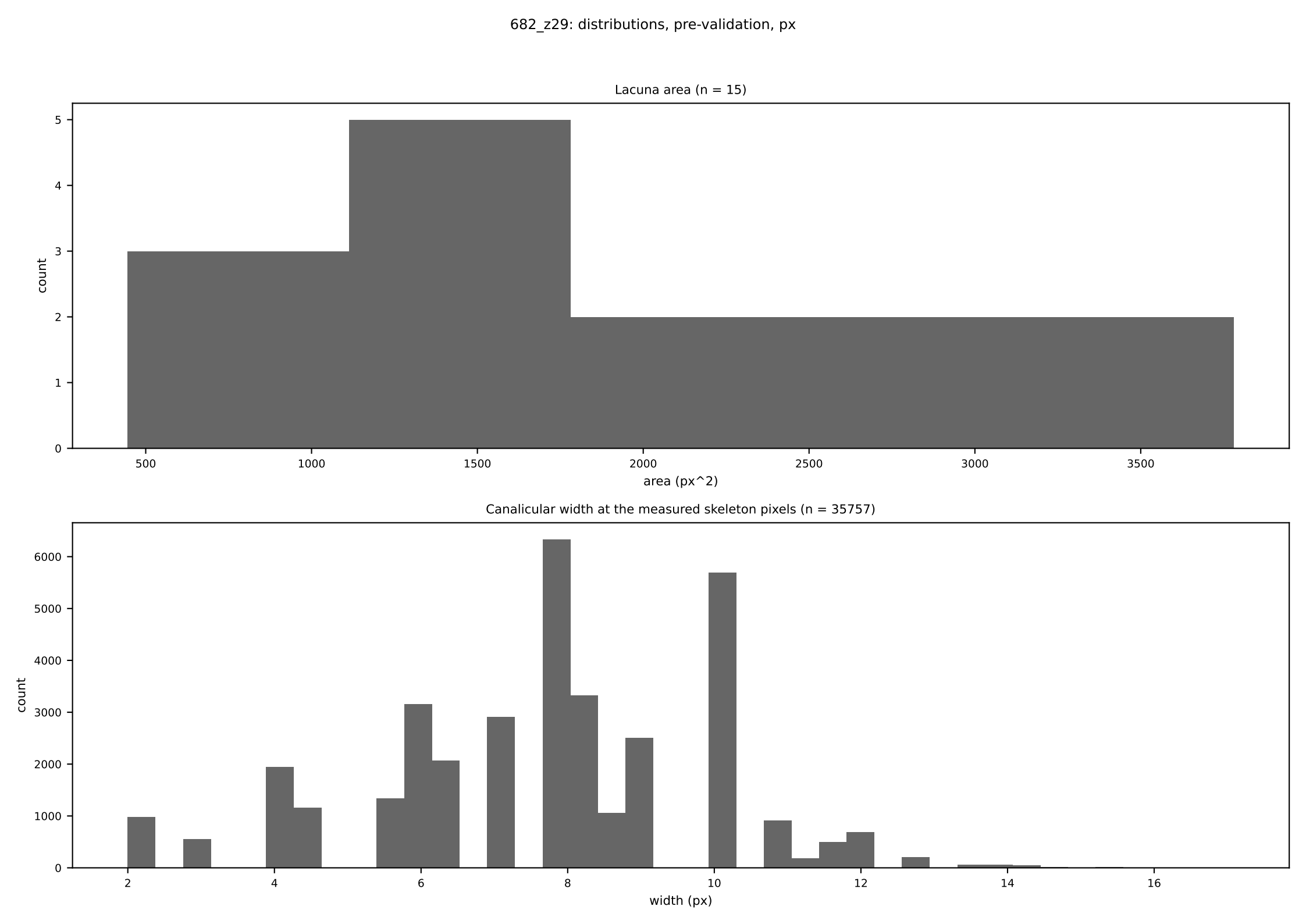
5	11	460	996	3959.72	164	24.1447	6.2493
6	10	481	950	3368.16	148	22.7579	6.9706
7	11	392	821	1236.62	60	20.6103	7.4798
8	8	376	951	1458.23	62	23.5198	6.2347
9	9	222	673	2846.64	116	24.54	6.8502
10	6	188	481	513.162	24	21.3817	6.6465
11	5	191	480	626.831	24	26.1179	8.6113
12	4	210	648	1494.79	58	25.7723	6.5206
13	11	445	938	1579.88	62	25.482	6.7132
14	10	272	452	539.688	21	25.6994	7.0847
15	2	245	624	854.541	39	21.9113	7.0877

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	262.096	5607	13819	0.5809	0.5584	0.0342429	0.0259787	1.5262
2	229.522	9151	23871	1	0.999	0.0607584	0.0562607	6.0996
3	120.811	6134	17911	1	1	0.0591783	0.0525934	4.9664
4	309.279	11331	27269	1	1	0.0405083	0.0452895	3.5567
5	235.238	9387	21888	1	1	0.0490039	0.0455044	4.6761
6	214.167	8637	19737	1	1	0.0556906	0.048133	4.6693
7	322.836	9940	21386	0.8402	0.7724	0.0394366	0.0383896	3.4073
8	229.48	8595	22042	1	1	0.0437464	0.0431449	3.4861
9	156.853	6922	19429	1	1	0.0320717	0.0346389	5.7379
10	109.498	4245	11066	1	0.9865	0.0442874	0.0434665	5.4796
11	161.817	5491	13376	1	1	0.0347842	0.0358852	3.0899
12	147.196	6914	19480	1	1	0.0303732	0.0332649	2.7175
13	254.693	9728	23664	1	1	0.0457442	0.0396383	4.3189
14	202.693	7711	16371	0.9376	0.8055	0.0352743	0.0276098	4.9336
15	47.4558	4187	11845	1	1	0.0585145	0.0526805	4.2144

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	85	150	219.51	395.664	2	6	6
2	548	1291	617.718	1522.9	12	16	18
3	319	942	410.635	1048.9	8	11	11
4	428	1116	519.818	1392.83	12	13	16
5	432	831	505.534	1101.01	12	15	14
6	429	900	539.99	1066.98	10	13	15
7	379	769	440.563	924.896	10	12	12
8	358	739	420.735	1071.29	10	12	12
9	199	515	263.522	786.423	5	7	10
10	176	392	210.338	535.576	6	6	6
11	191	450	221.823	534.09	5	7	8
12	157	481	236.48	752.21	6	7	8
13	388	744	500.718	1051.44	11	17	17
14	253	366	307.522	508.86	6	7	7
15	177	460	273.409	706.529	4	6	10


682_z29: distributions, pre-validation, px
Lacuna area (n = 15)
count
area (px^2)
0
1
2
3
4
5
500
1000
1500
2000
2500
3000
3500
Canalicular width at the measured skeleton pixels (n = 35757)
count
width (px)
0
1000
2000
3000
4000
5000
6000
2
4
6
8
10
12
14
16